

[**N.B.** – The figures of the right margin indicate full marks. Read the stems carefully and answer the associated questions. Answer any **FIVE** questions taking at least two questions from each group]

Group - A

1. **A and B are two events where $P(A) = \frac{1}{3}$, $P(B) = \frac{3}{4}$, and $P(A \cup B) = \frac{10}{12}$. A professor said that $P(A|B) = P(B|A)$.**

(a) What is the minimum value of probability?

(b) Why can't probability be negative? Explain with an example or theoretically.

(c) Find $P(A \cap B)$ from the stem.

(d) Evaluate the professor's assertion.
2. **A box contains 5 green, 8 yellow, and 7 orange marbles. A child takes out three balls at random.**

(a) What is a simple event?

(b) Prove using Venn diagram $P(A \cap \bar{B}) = P(A) - P(A \cap B)$

(c) What is the probability that two marbles are yellow and one green?

(d) Determine the probability that the marbles are the same color.
3. **A company wants to study the relationship between its advertising expenditure (in thousand \$) and monthly sales (in thousand \$). Data for 6 months are given below:**

Advertising (X, \$'000)	2	4	6	8	10	12
Sales (Y, \$'000)	48	56	62	70	78	84

- (a) What is correlation?

(b) Make a scatter plot from the given data and propose the association of the variables.

(c) Find and interpret the correlation coefficient.

(d) Fit the regression line and predict the monthly sales when advertising expenditure is \$9,000.

Group - B

4. **The joint probability mass function of two random variables X and Y is given by:**

$$P(X,Y) = \frac{3x+y}{48}; \quad x = 1, 2; \quad y = 0, 1, 2, 3$$

- (a) What is a discrete random variable?

(b) If $a < b$, what is the value of $F(b) - F(a)$, where F is cumulative distribution function?

(c) Find $P(X)$.

(d) Calculate $P(X|Y)$ and $P(X|Y = 1)$.
5. **The first four moments around 2 of a dataset were the following:**

-1, 5, 20, 90

- (a) What is raw moment?

(b) What is the standard deviation of the data in the stem?

(c) Determine the third central moment.

(d) Comment on the kurtosis of the given data. 4

6. The probability density function of a continuous random variable is

$$f(x) = \begin{cases} kx(x - 1), & 1 \leq x \leq 4 \\ 0, & otherwise \end{cases}$$

- (a) What is the range of probability? 1
- (b) Find the value of k 2
- (c) Find the Expectation of the function. 3
- (d) Find $V(3X+2)$. 4